

Limits, Continuity, and Differentiation of Real Functions

A Semi-Paper-Style Report on the Foundations of One-Variable Calculus

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Abstract

This report discusses the foundations of one-variable calculus through the concepts of limits, continuity, and differentiation. The aim is not merely to list computational rules, but to explain how these rules arise from precise definitions. The central object is a real-valued function of one real variable, and the main method is the logical progression from definition to proposition, proof, example, and interpretation. First, the epsilon-delta definition of a limit is introduced as a way of describing local behavior near a point. Next, continuity is treated as the compatibility between limiting behavior and function values. Finally, differentiation is presented as a refined limiting process that measures the best linear approximation of a function near a point. The report emphasizes that limits provide the language, continuity gives stability, and differentiation captures local change.

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1 Introduction

Calculus often begins with practical procedures: computing limits, checking continuity, and differentiating functions. However, these procedures are not independent tricks. They are connected by a common idea: the behavior of a function near a point can be described by comparing values of the function at nearby inputs. The concept of a limit formalizes this comparison. Continuity is the special case in which the limiting behavior agrees with the actual value of the function. Differentiation is a further refinement in which the change of a function is compared to the change of the input.

The purpose of this report is to organize these ideas in a semi-paper-style format. Each major section follows the order

definition $\longrightarrow$ proposition $\longrightarrow$ proof $\longrightarrow$ example $\longrightarrow$ interpretation.

This order is important. A definition introduces a concept, a proposition states a consequence, a proof explains why the consequence follows, and an example shows how the abstract statement works in a concrete situation. The interpretive viewpoint then records the meaning of the result in ordinary mathematical language.

The level of this report is intended for first- or second-year undergraduate students. Therefore, the discussion is rigorous but not unnecessarily abstract. The main setting is real functions defined on subsets of $\mathbb{R}$. Standard results such as the algebra of limits, the relation between continuity and limits, the intermediate value theorem, differentiability implying continuity, the rules of differentiation, and the mean value theorem are treated as core structural facts.

2 Functions and Local Behavior

Before defining limits, it is necessary to clarify what kind of object a function is and what it means to discuss the behavior of a function near a point. A limit at a point does not necessarily depend on the value of the function at that point. It depends on values of the function arbitrarily close to that point.

Definition 2.1 (Real-valued function). Let $D \subset \mathbb{R}$. A real-valued function on D is a rule

$$f : D \rightarrow \mathbb{R}$$

that assigns to each $x \in D$ exactly one real number $f(x)$. The set D is called the domain of f .

Definition 2.2 (Accumulation point). Let $D \subset \mathbb{R}$, and let $a \in \mathbb{R}$. The point a is called an accumulation point of D if, for every $\delta > 0$, there exists $x \in D$ such that

$$0 < |x - a| < \delta.$$

In other words, points of D distinct from a can be found arbitrarily close to a .

Proposition 2.3 (A limit concerns nearby values). *Let $f : D \rightarrow \mathbb{R}$, and let a be an accumulation point of D . If two functions $f, g : D \rightarrow \mathbb{R}$ satisfy*

$$f(x) = g(x)$$

for all $x \in D$ with $0 < |x - a| < r$ for some $r > 0$, then f and g have the same limit at a , whenever either limit exists.

Proof. Assume that $\lim_{x \rightarrow a} f(x) = L$. Let $\varepsilon > 0$ be given. By the definition of the limit, there exists $\delta_1 > 0$ such that, whenever $x \in D$ and

$$0 < |x - a| < \delta_1,$$

we have

$$|f(x) - L| < \varepsilon.$$

Set

$$\delta = \min\{\delta_1, r\}.$$

If $x \in D$ and $0 < |x - a| < \delta$, then $0 < |x - a| < r$, so $g(x) = f(x)$. Hence

$$|g(x) - L| = |f(x) - L| < \varepsilon.$$

Therefore $\lim_{x \rightarrow a} g(x) = L$. The converse follows by the same argument with f and g interchanged. $\square$

Example 2.4 (Changing the value at one point). Define

$$f(x) = x^2$$

for all real x , and define

$$g(x) = \begin{cases} x^2, & x \neq 1, \\ 100, & x = 1. \end{cases}$$

Then f and g agree near 1 except at $x = 1$ itself. Since

$$\lim_{x \rightarrow 1} f(x) = 1,$$

the proposition implies that

$$\lim_{x \rightarrow 1} g(x) = 1.$$

The value $g(1) = 100$ does not affect the limit.

Interpretive viewpoint. A limit is not a statement about what happens exactly at the point. It is a statement about what happens as one approaches the point. This is why the condition $0 < |x - a|$ appears in the definition. The exclusion of $x = a$ is not a technical accident; it expresses the basic meaning of limiting behavior.

3 Limits of Functions

The limit of a function is the foundation on which continuity and differentiation are built. The epsilon-delta definition may appear complicated at first, but its purpose is simple: it translates the phrase “ $f(x)$ becomes close to L when x becomes close to a ” into a precise mathematical condition.

Definition 3.1 (Limit of a function). Let $f : D \rightarrow \mathbb{R}$, and let a be an accumulation point of D . We say that $f(x)$ tends to L as x tends to a , and write

$$\lim_{x \rightarrow a} f(x) = L,$$

if for every $\varepsilon > 0$, there exists $\delta > 0$ such that, for every $x \in D$,

$$0 < |x - a| < \delta \implies |f(x) - L| < \varepsilon.$$

Proposition 3.2 (Uniqueness of limits). *If*

$$\lim_{x \rightarrow a} f(x) = L \quad \text{and} \quad \lim_{x \rightarrow a} f(x) = M,$$

then $L = M$.

Proof. Suppose, for contradiction, that $L \neq M$. Set

$$\varepsilon = \frac{|L - M|}{3} > 0.$$

Since $\lim_{x \rightarrow a} f(x) = L$, there exists $\delta_1 > 0$ such that

$$0 < |x - a| < \delta_1 \implies |f(x) - L| < \varepsilon.$$

Since $\lim_{x \rightarrow a} f(x) = M$, there exists $\delta_2 > 0$ such that

$$0 < |x - a| < \delta_2 \implies |f(x) - M| < \varepsilon.$$

Let $\delta = \min\{\delta_1, \delta_2\}$. Because a is an accumulation point of D , there exists $x \in D$ such that

$$0 < |x - a| < \delta.$$

For this x , the triangle inequality gives

$$|L - M| \leq |L - f(x)| + |f(x) - M| < \varepsilon + \varepsilon = 2\varepsilon = \frac{2}{3}|L - M|.$$

This is impossible. Therefore $L = M$. □

Example 3.3 (A basic epsilon-delta proof). We show that

$$\lim_{x \rightarrow 2} (3x + 1) = 7.$$

Let $\varepsilon > 0$ be given. We need $|(3x + 1) - 7| < \varepsilon$. Since

$$|(3x + 1) - 7| = |3x - 6| = 3|x - 2|,$$

it is enough to require

$$|x - 2| < \frac{\varepsilon}{3}.$$

Thus choose $\delta = \varepsilon/3$. Then

$$0 < |x - 2| < \delta \implies |(3x + 1) - 7| < \varepsilon.$$

Therefore $\lim_{x \rightarrow 2} (3x + 1) = 7$.

Interpretive viewpoint. The epsilon-delta definition is a game of control. The number ε describes the desired accuracy in the output, and δ describes how accurately the input must be controlled. A proof of a limit shows that every requested output accuracy can be achieved by choosing a sufficiently small input range.

4 Algebra of Limits

Many practical limit computations rely on algebraic rules. These rules are not separate from the epsilon-delta definition; rather, they are consequences of it. The algebra of limits explains why complicated limits can often be reduced to simpler ones.

Definition 4.1 (Pointwise algebra of functions). Let $f, g : D \rightarrow \mathbb{R}$, and let $c \in \mathbb{R}$. The functions $f + g$, cf , and fg are defined by

$$(f + g)(x) = f(x) + g(x), \quad (cf)(x) = cf(x), \quad (fg)(x) = f(x)g(x).$$

If $g(x) \neq 0$ for all $x \in D$, the quotient f/g is defined by

$$\left(\frac{f}{g}\right)(x) = \frac{f(x)}{g(x)}.$$

Proposition 4.2 (Algebra of limits). *Suppose that*

$$\lim_{x \rightarrow a} f(x) = L, \quad \lim_{x \rightarrow a} g(x) = M.$$

Then

$$\begin{aligned} \lim_{x \rightarrow a} (f(x) + g(x)) &= L + M, \\ \lim_{x \rightarrow a} cf(x) &= cL, \\ \lim_{x \rightarrow a} f(x)g(x) &= LM. \end{aligned}$$

If $M \neq 0$ and $g(x) \neq 0$ near a , then

$$\lim_{x \rightarrow a} \frac{f(x)}{g(x)} = \frac{L}{M}.$$

Proof. We prove the sum and product rules. The scalar rule is similar, and the quotient rule follows by combining the product rule with the reciprocal rule.

For the sum rule, let $\varepsilon > 0$. Since $f(x) \rightarrow L$, there exists $\delta_1 > 0$ such that

$$0 < |x - a| < \delta_1 \implies |f(x) - L| < \frac{\varepsilon}{2}.$$

Since $g(x) \rightarrow M$, there exists $\delta_2 > 0$ such that

$$0 < |x - a| < \delta_2 \implies |g(x) - M| < \frac{\varepsilon}{2}.$$

Let $\delta = \min\{\delta_1, \delta_2\}$. Then

$$|(f(x) + g(x)) - (L + M)| \leq |f(x) - L| + |g(x) - M| < \varepsilon.$$

This proves the sum rule.

For the product rule, write

$$f(x)g(x) - LM = f(x)(g(x) - M) + M(f(x) - L).$$

Since $f(x) \rightarrow L$, the function f is bounded near a . More precisely, there exists $\delta_0 > 0$ such that

$$0 < |x - a| < \delta_0 \implies |f(x) - L| < 1,$$

so $|f(x)| \leq |L| + 1$ near a . Let $B = |L| + 1$. Choose $\delta_1 > 0$ so that

$$|g(x) - M| < \frac{\varepsilon}{2B}$$

near a , and choose $\delta_2 > 0$ so that

$$|f(x) - L| < \frac{\varepsilon}{2(|M| + 1)}$$

near a . With $\delta = \min\{\delta_0, \delta_1, \delta_2\}$, we obtain

$$|f(x)g(x) - LM| \leq |f(x)| |g(x) - M| + |M| |f(x) - L| < \frac{\varepsilon}{2} + \frac{\varepsilon}{2} = \varepsilon.$$

Thus $f(x)g(x) \rightarrow LM$. □

Example 4.3 (Polynomial limits). Let

$$p(x) = 2x^3 - x + 5.$$

Using the algebra of limits,

$$\lim_{x \rightarrow 1} p(x) = 2 \cdot 1^3 - 1 + 5 = 6.$$

This computation is justified because polynomial functions are constructed from constants, the identity function, sums, and products.

Interpretive viewpoint. The algebra of limits is what makes ordinary limit computation possible. Without these rules, every polynomial or rational limit would require a direct epsilon-delta proof. The rules show that once the behavior of simple functions is known, the behavior of functions built from them follows automatically.

5 Continuity

Continuity connects the limiting behavior of a function with its value. A function is continuous at a point when approaching the point through the domain forces the function values to approach the function value at that point.

Definition 5.1 (Continuity at a point). Let $f : D \rightarrow \mathbb{R}$, and let $a \in D$. The function f is continuous at a if for every $\varepsilon > 0$, there exists $\delta > 0$ such that, for every $x \in D$,

$$|x - a| < \delta \implies |f(x) - f(a)| < \varepsilon.$$

If f is continuous at every point of D , then f is called continuous on D .

Proposition 5.2 (Continuity and limits). Let $f : D \rightarrow \mathbb{R}$, and let $a \in D$ be an accumulation point of D . Then f is continuous at a if and only if

$$\lim_{x \rightarrow a} f(x) = f(a).$$

Proof. Assume first that f is continuous at a . Let $\varepsilon > 0$. By continuity, there exists $\delta > 0$ such that

$$|x - a| < \delta \implies |f(x) - f(a)| < \varepsilon.$$

In particular, the same implication holds whenever $0 < |x - a| < \delta$. Hence

$$\lim_{x \rightarrow a} f(x) = f(a).$$

Conversely, assume that $\lim_{x \rightarrow a} f(x) = f(a)$. Let $\varepsilon > 0$. Then there exists $\delta_1 > 0$ such that

$$0 < |x - a| < \delta_1 \implies |f(x) - f(a)| < \varepsilon.$$

To obtain continuity, we must also allow $x = a$. But if $x = a$, then

$$|f(x) - f(a)| = 0 < \varepsilon.$$

Therefore the same conclusion holds for all $x \in D$ satisfying $|x - a| < \delta_1$. Thus f is continuous at a . $\square$

Example 5.3 (A removable discontinuity). Define

$$f(x) = \begin{cases} \frac{x^2 - 1}{x - 1}, & x \neq 1, \\ 5, & x = 1. \end{cases}$$

For $x \neq 1$,

$$\frac{x^2 - 1}{x - 1} = x + 1.$$

Hence

$$\lim_{x \rightarrow 1} f(x) = \lim_{x \rightarrow 1} (x + 1) = 2.$$

However, $f(1) = 5$. Therefore

$$\lim_{x \rightarrow 1} f(x) \neq f(1),$$

so f is not continuous at 1.

Interpretive viewpoint. Continuity says that the function value is consistent with the surrounding values. A discontinuity occurs when this consistency fails. In a removable discontinuity, the surrounding behavior is well-defined, but the assigned value at the point is wrong or missing.

6 Sequential Characterization of Continuity

The epsilon-delta definition gives direct control of input and output. Another useful viewpoint is sequential: a function is continuous if it preserves limits of convergent sequences. This connects calculus with the structure of the real line.

Definition 6.1 (Convergence of a sequence). A sequence (x_n) of real numbers converges to $a \in \mathbb{R}$ if for every $\varepsilon > 0$, there exists $N \in \mathbb{N}$ such that

$$n \geq N \implies |x_n - a| < \varepsilon.$$

In this case, we write $x_n \rightarrow a$.

Proposition 6.2 (Sequential characterization of continuity). *Let $f : D \rightarrow \mathbb{R}$, and let $a \in D$. Then f is continuous at a if and only if for every sequence (x_n) in D satisfying $x_n \rightarrow a$, we have*

$$f(x_n) \rightarrow f(a).$$

Proof. Assume that f is continuous at a , and let (x_n) be a sequence in D with $x_n \rightarrow a$. Let $\varepsilon > 0$. By continuity, there exists $\delta > 0$ such that

$$|x - a| < \delta \implies |f(x) - f(a)| < \varepsilon.$$

Since $x_n \rightarrow a$, there exists $N \in \mathbb{N}$ such that $n \geq N$ implies $|x_n - a| < \delta$. Hence $n \geq N$ implies

$$|f(x_n) - f(a)| < \varepsilon.$$

Therefore $f(x_n) \rightarrow f(a)$.

Conversely, suppose that the sequential condition holds, but f is not continuous at a . Then there exists $\varepsilon_0 > 0$ such that for every $\delta > 0$, one can find $x \in D$ with

$$|x - a| < \delta \quad \text{and} \quad |f(x) - f(a)| \geq \varepsilon_0.$$

For each $n \in \mathbb{N}$, choose $x_n \in D$ with

$$|x_n - a| < \frac{1}{n} \quad \text{and} \quad |f(x_n) - f(a)| \geq \varepsilon_0.$$

Then $x_n \rightarrow a$, but $f(x_n)$ does not converge to $f(a)$. This contradicts the sequential condition. Hence f is continuous at a . $\square$

Example 6.3 (Detecting discontinuity by a sequence). Let

$$f(x) = \begin{cases} 1, & x > 0, \\ 0, & x \leq 0. \end{cases}$$

Take $x_n = 1/n$. Then $x_n \rightarrow 0$, but

$$f(x_n) = 1$$

for every n , while $f(0) = 0$. Thus $f(x_n) \not\rightarrow f(0)$, and f is not continuous at 0.

Interpretive viewpoint. The sequential characterization is often easier to use when disproving continuity. To show that a function is not continuous, it is enough to find a sequence approaching the point whose function values fail to approach the expected value.

7 The Intermediate Value Property

Continuity is not only a local condition. On intervals, it also has global consequences. The intermediate value theorem expresses the idea that a continuous function cannot jump over a value.

Definition 7.1 (Interval). A subset $I \subset \mathbb{R}$ is called an interval if, whenever $x, y \in I$ and $x < z < y$, we also have $z \in I$. In other words, an interval contains every point between any two of its points.

Theorem 7.2 (Intermediate value theorem). Let $f : [a, b] \rightarrow \mathbb{R}$ be continuous. If N is a real number between $f(a)$ and $f(b)$, then there exists $c \in [a, b]$ such that

$$f(c) = N.$$

Proof. Assume, for simplicity, that $f(a) < N < f(b)$. Define

$$S = \{x \in [a, b] : f(x) < N\}.$$

The set S is nonempty because $a \in S$, and it is bounded above by b . By the completeness property of $\mathbb{R}$, the supremum

$$c = \sup S$$

exists.

We show that $f(c) = N$. If $f(c) < N$, then by continuity at c , values of $f(x)$ remain less than N for x sufficiently close to c . This would allow a point greater than c to belong to S , contradicting that c is an upper bound of S . If $f(c) > N$, then by continuity at c , values of $f(x)$ remain greater than N for x sufficiently close to c . This contradicts the fact that points of S can be found arbitrarily close to c from the left. Therefore neither $f(c) < N$ nor $f(c) > N$ is possible, so $f(c) = N$. $\square$

Example 7.3 (Existence of a root). Consider

$$f(x) = x^3 + x - 1$$

on $[0, 1]$. Since f is a polynomial, it is continuous. Moreover,

$$f(0) = -1, \quad f(1) = 1.$$

Since 0 lies between $f(0)$ and $f(1)$, the intermediate value theorem implies that there exists $c \in [0, 1]$ such that

$$c^3 + c - 1 = 0.$$

Interpretive viewpoint. The intermediate value theorem converts continuity into an existence theorem. It may not give the exact value of c , but it proves that such a value must exist. This is one of the first places where rigorous analysis shows a result that is not merely computational.

8 Differentiation

Differentiation measures the local rate of change of a function. Unlike continuity, which compares $f(x)$ with $f(a)$, differentiation compares the quotient of changes

$$\frac{f(x) - f(a)}{x - a}$$

with a limiting number.

Definition 8.1 (Derivative at a point). Let $f : D \rightarrow \mathbb{R}$, and suppose that $a \in D$ is an accumulation point of D . If the limit

$$\lim_{x \rightarrow a} \frac{f(x) - f(a)}{x - a}$$

exists, then f is said to be differentiable at a . The value of this limit is called the derivative of f at a , and is denoted by

$$f'(a) = \lim_{x \rightarrow a} \frac{f(x) - f(a)}{x - a}.$$

Proposition 8.2 (Differentiability implies continuity). *If f is differentiable at a , then f is continuous at a .*

Proof. Assume that f is differentiable at a . Then the limit

$$\lim_{x \rightarrow a} \frac{f(x) - f(a)}{x - a} = f'(a)$$

exists. For $x \neq a$, write

$$f(x) - f(a) = \frac{f(x) - f(a)}{x - a} (x - a).$$

Taking limits as $x \rightarrow a$, we obtain

$$\lim_{x \rightarrow a} (f(x) - f(a)) = \left(\lim_{x \rightarrow a} \frac{f(x) - f(a)}{x - a} \right) \left(\lim_{x \rightarrow a} (x - a) \right) = f'(a) \cdot 0 = 0.$$

Thus

$$\lim_{x \rightarrow a} f(x) = f(a),$$

so f is continuous at a . □

Example 8.3 (A continuous function that is not differentiable). Let

$$f(x) = |x|.$$

Then f is continuous at 0, because $|x| \rightarrow 0$ as $x \rightarrow 0$. However,

$$\frac{f(x) - f(0)}{x - 0} = \frac{|x|}{x}.$$

If $x > 0$, this quotient equals 1. If $x < 0$, it equals -1 . The two one-sided limits are different, so the derivative at 0 does not exist.

Interpretive viewpoint. Differentiability is stronger than continuity. A differentiable function must be continuous, but a continuous function need not be differentiable. Geometrically, continuity means that the graph has no jump at the point, while differentiability requires a well-defined tangent slope.

9 Rules of Differentiation

The derivative is defined as a limit, but in practice derivatives are computed using rules. These rules are consequences of the definition and the algebra of limits.

Definition 9.1 (Derivative function). Let $f : D \rightarrow \mathbb{R}$. If f is differentiable at every point of a set $E \subset D$, then the function

$$f' : E \rightarrow \mathbb{R}, \quad x \mapsto f'(x)$$

is called the derivative function of f on E .

Proposition 9.2 (Sum, product, and chain rules). *Let f and g be differentiable at a , and let $c \in \mathbb{R}$. Then $f + g$, cf , and fg are differentiable at a , and*

$$(f + g)'(a) = f'(a) + g'(a),$$

$$(cf)'(a) = cf'(a),$$

$$(fg)'(a) = f'(a)g(a) + f(a)g'(a).$$

If g is differentiable at a , h is differentiable at $g(a)$, and the composition $h \circ g$ is defined near a , then

$$(h \circ g)'(a) = h'(g(a))g'(a).$$

Proof. The sum and scalar rules follow immediately from the definition of the derivative and the algebra of limits. We prove the product rule. For $x \neq a$,

$$\frac{f(x)g(x) - f(a)g(a)}{x - a} = \frac{f(x)g(x) - f(a)g(x) + f(a)g(x) - f(a)g(a)}{x - a}.$$

Thus

$$\frac{f(x)g(x) - f(a)g(a)}{x - a} = g(x)\frac{f(x) - f(a)}{x - a} + f(a)\frac{g(x) - g(a)}{x - a}.$$

Since differentiability implies continuity, $g(x) \rightarrow g(a)$ as $x \rightarrow a$. Therefore, by the algebra of limits,

$$(fg)'(a) = g(a)f'(a) + f(a)g'(a).$$

For the chain rule, write

$$h(y) - h(g(a)) = (h'(g(a)) + r(y))(y - g(a)),$$

where $r(y) \rightarrow 0$ as $y \rightarrow g(a)$. Put $y = g(x)$. Since g is differentiable at a , it is continuous at a , so $g(x) \rightarrow g(a)$. Hence

$$h(g(x)) - h(g(a)) = (h'(g(a)) + r(g(x)))(g(x) - g(a)).$$

Dividing by $x - a$ and taking the limit gives

$$(h \circ g)'(a) = h'(g(a))g'(a).$$

□

Example 9.3 (Using the rules). Let

$$F(x) = (x^2 + 1)^5.$$

Set $g(x) = x^2 + 1$ and $h(u) = u^5$. Then $F = h \circ g$. Since

$$g'(x) = 2x, \quad h'(u) = 5u^4,$$

the chain rule gives

$$F'(x) = 5(x^2 + 1)^4 \cdot 2x = 10x(x^2 + 1)^4.$$

Interpretive viewpoint. The differentiation rules are not independent formulas. They are condensed forms of limit computations. The product rule reflects the fact that both factors may change, and the chain rule reflects the propagation of a small change through a composition of functions.

10 The Mean Value Theorem

The mean value theorem is one of the central results connecting local and global behavior. It states that the average rate of change on an interval is achieved as an instantaneous rate of change at some point inside the interval.

Definition 10.1 (Local extremum). Let $f : D \rightarrow \mathbb{R}$, and let $c \in D$. The function f has a local maximum at c if there exists $r > 0$ such that

$$f(x) \leq f(c)$$

for all $x \in D$ with $|x - c| < r$. Similarly, f has a local minimum at c if there exists $r > 0$ such that

$$f(x) \geq f(c)$$

for all $x \in D$ with $|x - c| < r$.

Proposition 10.2 (Fermat's theorem). *If f has a local maximum or local minimum at an interior point c , and if f is differentiable at c , then*

$$f'(c) = 0.$$

Proof. Assume that f has a local maximum at c . Then for x sufficiently close to c ,

$$f(x) - f(c) \leq 0.$$

If $x > c$, then

$$\frac{f(x) - f(c)}{x - c} \leq 0.$$

Taking the right-hand limit gives $f'(c) \leq 0$. If $x < c$, then $x - c < 0$, so

$$\frac{f(x) - f(c)}{x - c} \geq 0.$$

Taking the left-hand limit gives $f'(c) \geq 0$. Since the derivative exists, the two one-sided limits agree, and therefore $f'(c) = 0$. The case of a local minimum is similar, with the inequalities reversed. $\square$

Example 10.3 (A stationary point). Let

$$f(x) = x^2.$$

The function has a local minimum at 0. Since $f'(x) = 2x$, we have

$$f'(0) = 0,$$

as Fermat's theorem predicts.

Interpretive viewpoint. Fermat's theorem explains why derivatives are useful in optimization. At an interior maximum or minimum, the graph cannot be increasing or decreasing to first order. Thus the tangent slope must vanish, provided the tangent slope exists.

Theorem 10.4 (Mean value theorem). *Let $f : [a, b] \rightarrow \mathbb{R}$ be continuous on $[a, b]$ and differentiable on (a, b) . Then there exists $c \in (a, b)$ such that*

$$f'(c) = \frac{f(b) - f(a)}{b - a}.$$

Proof. Define the auxiliary function

$$\phi(x) = f(x) - \left(\frac{f(b) - f(a)}{b - a} \right) (x - a).$$

Then ϕ is continuous on $[a, b]$ and differentiable on (a, b) . Moreover,

$$\phi(a) = f(a),$$

and

$$\phi(b) = f(b) - (f(b) - f(a)) = f(a).$$

Thus $\phi(a) = \phi(b)$. By Rolle's theorem, which follows from Fermat's theorem and the extreme value theorem, there exists $c \in (a, b)$ such that

$$\phi'(c) = 0.$$

Since

$$\phi'(x) = f'(x) - \frac{f(b) - f(a)}{b - a},$$

we obtain

$$f'(c) = \frac{f(b) - f(a)}{b - a}.$$

□

Example 10.5 (Applying the mean value theorem). Let $f(x) = x^2$ on $[1, 3]$. Then

$$\frac{f(3) - f(1)}{3 - 1} = \frac{9 - 1}{2} = 4.$$

Since $f'(x) = 2x$, the mean value theorem predicts a point $c \in (1, 3)$ such that

$$2c = 4.$$

Thus $c = 2$, which indeed belongs to $(1, 3)$.

Interpretive viewpoint. The mean value theorem bridges local and global information. The derivative is local, while the secant slope between a and b is global. The theorem says that under continuity and differentiability assumptions, the local behavior must match the global average somewhere.

11 Differentiation as Linear Approximation

A deeper interpretation of differentiation is that a differentiable function is locally approximated by a linear function. This viewpoint is especially important because it generalizes naturally to multivariable calculus and functional analysis.

Definition 11.1 (First-order approximation). Let f be differentiable at a . The first-order linear approximation of f at a is

$$L_a(x) = f(a) + f'(a)(x - a).$$

The error term is

$$R_a(x) = f(x) - L_a(x).$$

Proposition 11.2 (Derivative as best first-order approximation). *If f is differentiable at a , then*

$$f(x) = f(a) + f'(a)(x - a) + r(x)(x - a),$$

where

$$\lim_{x \rightarrow a} r(x) = 0.$$

Equivalently,

$$\lim_{x \rightarrow a} \frac{f(x) - f(a) - f'(a)(x - a)}{x - a} = 0.$$

Proof. For $x \neq a$, define

$$r(x) = \frac{f(x) - f(a)}{x - a} - f'(a).$$

Since $f'(a)$ is the limit of the difference quotient, we have

$$\lim_{x \rightarrow a} r(x) = 0.$$

Rearranging the definition of $r(x)$, we obtain

$$f(x) - f(a) = (f'(a) + r(x))(x - a),$$

which is the desired formula. □

Example 11.3 (Linear approximation of the square root). Let

$$f(x) = \sqrt{x}$$

near $a = 1$. Since

$$f(1) = 1, \quad f'(x) = \frac{1}{2\sqrt{x}}, \quad f'(1) = \frac{1}{2},$$

the first-order approximation is

$$L_1(x) = 1 + \frac{1}{2}(x - 1).$$

For $x = 1.04$, this gives

$$\sqrt{1.04} \approx 1 + \frac{1}{2}(0.04) = 1.02.$$

Interpretive viewpoint. Differentiation is not only about slopes. It is also about replacing a nonlinear function by the best linear model near a point. The smaller $|x - a|$ is, the less significant the error term becomes compared with $x - a$.

12 Overall Consideration

The concepts of limit, continuity, and differentiation form a hierarchy. Limits describe approach. Continuity requires that approach to agree with the actual function value. Differentiation requires the quotient of changes to approach a finite value. Thus each concept adds a stronger condition on the local behavior of a function.

From a logical point of view, limits are the most fundamental of the three. The definition of continuity is naturally expressed by a limit, and the definition of derivative is itself a limit of difference quotients. This explains why the study of calculus must eventually pass through the epsilon-delta definition. Without the concept of a limit, continuity and differentiation remain intuitive but not fully precise.

The examples also show that the implications are one-directional. Differentiability implies continuity, but continuity does not imply differentiability. Continuity at a point implies the

existence of the corresponding limit equal to the function value, but a limit can exist even when the function is not continuous at that point. These distinctions prevent common misunderstandings. A function can have a limit at a point without being continuous there, and it can be continuous without having a derivative.

The deeper meaning is that calculus is a theory of local structure. Limits describe local tendency, continuity describes local stability, and differentiation describes local linearity. This viewpoint prepares the way for later subjects such as multivariable calculus, differential equations, metric spaces, and functional analysis, where the same ideas appear in more abstract forms.

13 Conclusion

This report has examined limits, continuity, and differentiation of real functions in a structured way. The epsilon-delta definition of limit provides the precise foundation for discussing local behavior. Continuity is then understood as the compatibility of a function value with its limiting behavior. Differentiation is a further refinement that measures the limiting ratio of output change to input change.

The main results discussed here show how these ideas interact. The uniqueness and algebra of limits make limit computations reliable. The characterization of continuity by limits and sequences clarifies the meaning of continuous behavior. The intermediate value theorem shows that continuity has powerful global consequences on intervals. Differentiability implies continuity, and the rules of differentiation follow from the limit definition. Finally, the mean value theorem and the linear approximation viewpoint show that derivatives connect local rates of change with global behavior and first-order models.

Therefore, limits, continuity, and differentiation should not be viewed as separate chapters of calculus. They form a single conceptual chain. Limits provide the language, continuity gives stability, and differentiation reveals the local linear structure of functions.

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